🧪Hands-On Lab Exercise: Introduction to Shell Scripting

Author: Dr. Sandeep Kumar Sharma
Topic: Getting Started with Shell Script
Lab Type: Practical + Conceptual Learning

📘What is a Shell Script?

A **shell script** is a file that contains a group of Linux commands written in sequence.
These commands are executed one by one by the system using a **shell** (like bash).

In simple words:

> A shell script is a way to tell the computer what to do, step-by-step, using commands.

🕯️ Why Do We Need Shell Scripts?

We use shell scripts because: - To **automate repeated tasks** - To **save time** - To **reduce manual work** - To **avoid human mistakes** - To **run multiple commands together**

Example: Instead of typing 10 commands daily, we write them once in a script and run it with one command.

🕖Purpose of Using Shell Scripts

Shell scripts help us to: - Automate system tasks - Manage users and files - Take backups - Monitor systems - Start/stop services - Deploy applications - Configure servers

In real life IT work, shell scripts are used in: - DevOps - Cloud - System Administration - Automation - CI/CD pipelines

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📊Learning Objective

After this lab, participants will be able to: - Understand what a shell script is - Understand why shell scripts are used - Understand the purpose of shell scripting - Create their first shell script - Run a shell script - Understand each line of a script clearly

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🔗Learning Outcome

After completing this lab, participants will: - Know how to create a `.sh` file - Know how to write commands inside a script - Know how to give permission to a script - Know how to execute a script - Understand basic script structure - Build confidence to write simple automation scripts

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🧪Lab 1: Creating Your First Shell Script (Hello World)

Step 1: Open Terminal

Open your Linux terminal.

Step 2: Create a Script File

```
touch hello.sh
```

This command creates a file named `hello.sh`

Step 3: Open the File in Editor

```
nano hello.sh
```

Step 4: Write Your First Script

Paste this inside the file:

```
#!/bin/bash
echo "Hello World"
```

📖Explanation of Each Line

Line 1:

```
#!/bin/bash
```

This is called **shebang**

Meaning: - `#!` → tells the system this is a script - `/bin/bash` → tells the system to use **bash shell** to run this script

In simple words:

> "Run this file using bash shell"

Line 2:

```
echo "Hello World"
```

- `echo` → command to print output on screen
- `"Hello World"` → the text that will be displayed

Output will be:

```
Hello World
```

🔐Step 5: Give Execute Permission

```
chmod +x hello.sh
```

Meaning: This command gives permission to run the script.

📩Step 6: Run the Script

```
./hello.sh
```

📤Output

```
Hello World
```

Understanding the Flow

1. You created a file
2. You wrote commands inside it
3. You gave permission
4. You executed it
5. System ran commands one-by-one

🧰Real-Life Meaning

This script is small, but: - Same method is used for big automation - Same structure is used in DevOps - Same concept is used in production systems

Small scripts → Big automation systems

📌Summary

- Shell script = file with Linux commands
- Used for automation
- Saves time
- Reduces errors
- Improves productivity
- Makes system work easier

💴End of Lab

This is your **first step into automation world** 🚀

More complex scripts will use: - Variables - Conditions - Loops - Functions - Inputs - Arguments

But the foundation starts here.

Author:
Dr. Sandeep Kumar Sharma
Trainer | Mentor | DevOps | Linux | Automation